Final Examination: DSAI6707 Advanced Computational Intelligence (Sample Questions)

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1 Multiple-Choice Questions (30 Marks)

Instructions: This section consists of 10 questions. Each question carries **3 marks**. For each question, please select the **single best answer** from the choices given. Only one answer is correct for each question. Mark your answers clearly in the examination booklet.

1. Which of the following fundamental benchmark problems cannot be classified correctly by Frank Rosenblatt's single-layer perceptron due to the lack of linear separability?
   (a) Two-input logical AND
   (b) Two-input logical OR
   (c) Two-input logical NAND
   (d) Two-input logical XOR (Exclusive-OR)

2. Consider the representational capacity of single-layer perceptron, two-layer MLP and three-layer MLP (with non-linear differentiable activation). Which comparison is correct?
   (a) Single-layer perceptron: arbitrary decision regions; two-layer MLP: half-plane; three-layer MLP: convex regions
   (b) Single-layer perceptron: half-plane (linear separation); two-layer MLP: convex regions; three-layer MLP: arbitrarily complex decision regions (bounded by neuron count)
   (c) All three architectures have identical expressive power given sufficient training epochs
   (d) XOR problem can be solved by single-layer perceptron when using logistic activation function

2 Part II: Written-response Questions (70 Marks)

Instructions: This section consists of 6 computational and analytical problems. Please answer all questions and show your calculation steps and derivations clearly.

1. (10 Marks) Consider a one-dimensional linear regression problem where the model hypothesis is defined as:

$$\hat{y} = wx + b. \tag{1}$$

The objective is to minimize the Mean Squared Error (MSE) loss function over the dataset:

$$J(w, b) = \frac{1}{2m} \sum_{i=1}^{m} (\hat{y}^{(i)} - y^{(i)})^2. \tag{2}$$

For an individual training sample $(x^{(i)}, y^{(i)})$, the instantaneous loss is defined as:

$$L^{(i)}(w, b) = \frac{1}{2}(wx^{(i)} + b - y^{(i)})^2. \tag{3}$$

You are given a small training dataset consisting of $m = 3$ samples: $\mathcal{D} = \{(1, 2), (2, 3), (3, 5)\}$.

Questions:

(1) (3 Marks) Analytical Closed-Form Solution (Normal Equation): Formulate the design matrix X and target vector y. Compute the optimal parameter vector $\theta^* = [b^*, w^*]^T$ directly using the Normal Equation:

$$\theta^* = (X^T X)^{-1} X^T y.$$

(2) (4 Marks) Iterative Optimization via Stochastic Gradient Descent (SGD): Suppose the model parameters are initialized to $w_0 = 0.0$ and $b_0 = 0.0$, with a fixed learning rate of $\eta = 0.1$. Train the model for one complete epoch using SGD, presenting the samples sequentially in the order:

$$(x^{(1)}, y^{(1)}) = (1, 2) \rightarrow (x^{(2)}, y^{(2)}) = (2, 3) \rightarrow (x^{(3)}, y^{(3)}) = (3, 5).$$

For each iteration step $t \in \{1, 2, 3\}$, explicitly compute:

– The predicted value $\hat{y}^{(i)}$.

– The prediction error $e = \hat{y}^{(i)} - y^{(i)}$.

– The updated parameters w_t and b_t.

(3) (3 Marks) Discussion & Comparison:

Briefly compare the computational time complexity of the Normal Equation versus Gradient Descent with respect to the number of features d.

2. (12 Marks) Consider a binary classification problem on a one-dimensional feature space with the following four training exemplars:

   Class 1 (Desired output $d = 1$): $x^{(1)} = 0.5, \quad x^{(2)} = 2.0$

   Class 2 (Desired output $d = 0$): $x^{(3)} = -1.0, \quad x^{(4)} = -2.0$

   To incorporate the bias parameter b, use the augmented input vector $\mathbf{x}' = [1, x]^T$ and the augmented weight vector $\mathbf{w}' = [b, w]^T$. The neuron utilizes the standard hard-limiter activation function:

$$y = f(v) = \begin{cases} 1, & \mathbf{w}'^T\mathbf{x}' \geq 0 \\ 0, & \mathbf{w}'^T\mathbf{x}' < 0 \end{cases} \tag{4}$$

   The perceptron weight vector is updated according to the Perceptron Learning Algorithm:

$$\mathbf{w}'_{new} = \mathbf{w}' + \eta(d - y)\mathbf{x}'.$$

   where η is the learning rate. Assume the initial weight vector is initialized to $\mathbf{w}'_0 = [-1.5, -2.0]^T$, and the learning rate is set to $\eta = 1.0$. The training exemplars are presented sequentially in the order: $x^{(1)} \to x^{(3)} \to x^{(2)} \to x^{(4)}$ to complete one training epoch.

   Questions:

   (1) (8 Marks) Step-by-step Execution of Epoch 1: Trace the perceptron learning algorithm across all 4 iterations in the first epoch. For each step $k \in \{1, 2, 3, 4\}$, explicitly compute: The augmented input vector $\mathbf{x}'$; The linear combiner output (induced local field) $v = \mathbf{w}'^T\mathbf{x}'$; The actual output y; The error signal $(d - y)$; and The updated weight vector $\mathbf{w}'_k$.

   (2) (4 Marks) Convergence and Decision Boundary: Determine whether the perceptron algorithm has converged after this single epoch. If so, verify the model on all four training instances and derive the final decision boundary (the specific value of x where the boundary lies).